GIT TO RENDER DEPLOYMENT CHEATSHEET

End-to-End Quick Reference: Local Changes -> GitHub -> Render Free Web Service

STEP 1: LOCAL GIT WORKFLOW & COMMITS

- **Check Working Tree:** Review changed and untracked files before committing.

```
git status
git add .
git commit -m "feat: updates for production deployment"
```

- **Branch Check:** Ensure you are working on the production deployment branch (usually main).

```
git branch -M main
git push origin main
```

STEP 2: GITHUB DESKTOP PUSH (GUI ALTERNATIVE)

- **Open Local Repo:** File -> Add Local Repository (Ctrl + O) -> Select your project root folder.
- **Commit Changes:** Enter summary in bottom-left box -> Click "Commit to main".
- **Push / Publish:** Click the top-bar button "Push origin" (or "Publish repository" if brand new).

STEP 3: DEPLOY VIA RENDER AUTOMATION SCRIPT

- **One-Command Deploy:** Provisions service, connects repo, and injects production env vars.

```
$env:RENDER_API_KEY="rnd_your_key_here"
node scripts/deploy-render.mjs
```

- **Script Logic:** Automatically attaches repo https://github.com/bskcode/TECHMATE to Render.

STEP 4: RENDER DASHBOARD CONFIGURATION (FREE TIER)

- **Create Web Service:** Dashboard -> New + -> Web Service -> Select repository.
- **Runtime & Build:** Runtime: Node | Region: Singapore | Plan: Free ($0/mo)

```
Build Command : npm install && npm run build
Start Command : npm run start
```

- **Essential Env Vars:** Set under the "Environment Variables" section of the service:

NODE_ENV	production	Production optimization mode
PORT	10000	Render internal traffic port
AI_PROVIDER	groq	Primary inference engine
GROQ_MODEL	llama-3.3-70b-versatile	Ultra-fast 70B model
GROQ_API_KEY	gsk_...	Your private Groq API key

STEP 5: LIVE VERIFICATION & FREE TIER NOTES

- **Health Check:** Visit https://<app-name>.onrender.com/api/health (returns 200 healthy).
- **Inactivity Sleep:** Free tier spins down after 15 min idle; first wake-up takes ~45s.
- **Auto-Deploy:** Render auto-redeploys whenever you push new commits to GitHub main.